

```

t.create
obj.imp.getAlternativeServices(t)
service := t[1,1][1,1]

part := @ --if part.id = 4 then debug
end
tab.create; allWorker := WorkerPartAssignment
if part /= void
    j := allWorker.getRowNo(part)
    if j = -1
        j := 1
        while j<= allWorker.yDim AND allWorker[0,j]/=void
            j := j + 1
        end
        if j<= allWorker.yDim
            allWorker[0,j] := part
            expObj := allWorker[1,j]
            tab.writeRow(1,1, expObj, service, 1 )
            broker.engage( obj, type, tab )
        end
    else
        expObj := allWorker[1,j]
        tab.writeRow(1,1, expObj, service, 1 )
        broker.engage( obj, type, tab )
    end
end
end

```

SimTalk

[engage \[SimTalk\]](#)

See also

[Importer Request Control \[Broker\]](#)

AGVPool

AGVPool

Use the object *AGVPool* for creating *automated guided vehicles* (AGVs). You can then model AGV systems in your installation in which the AGVs are not bound to a permanently installed route network.

Image

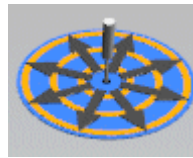


Description

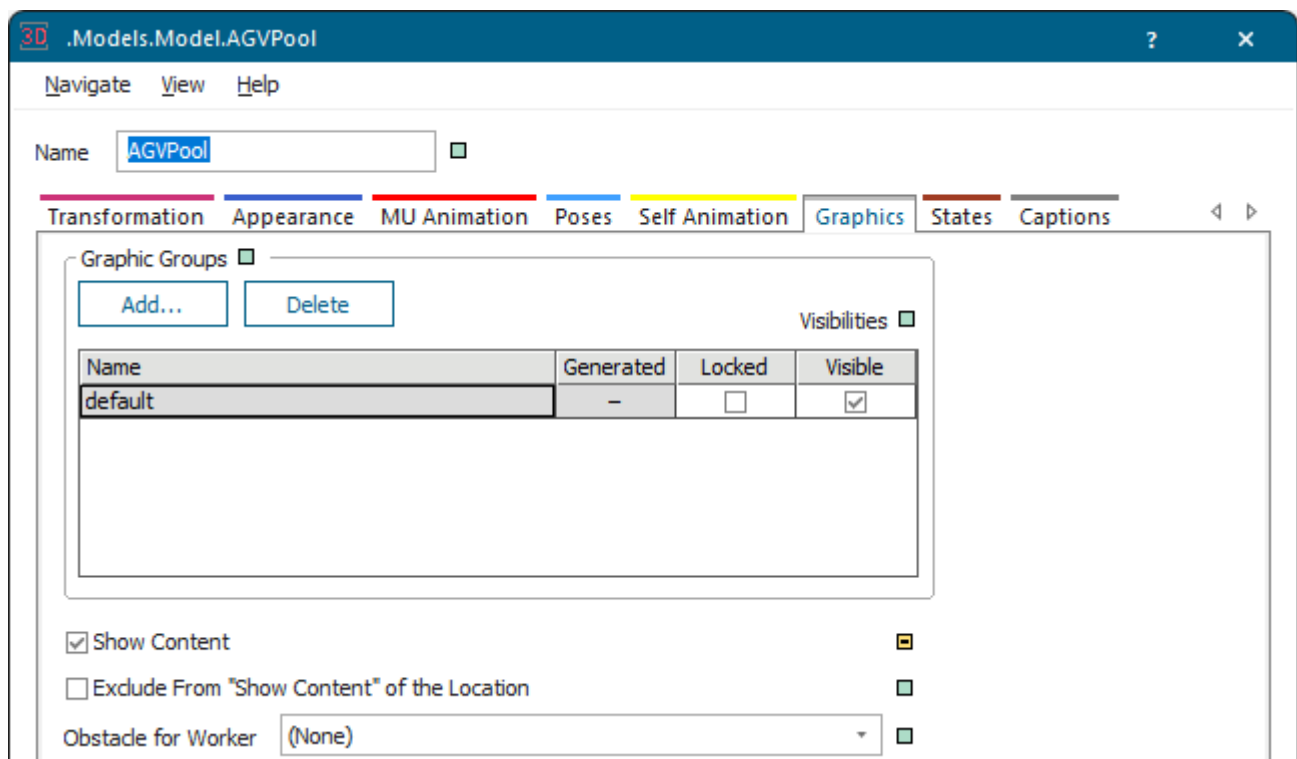
You might model the supervisor of an plant or the foreman of a shop with it. The *AGVPool* creates *automated guided vehicles*. You can then model AGV systems in your installation in which the AGVs are not bound to a permanently installed route network.

This is especially helpful if you need to produce smaller lot sizes with a high variant mix on a single production line. Instead of fixed production lines you can then use AGVs to transport the loaded products through a modular structured installation. The AGVs can, depending on the situation, create and change virtual tracks on which they drive.

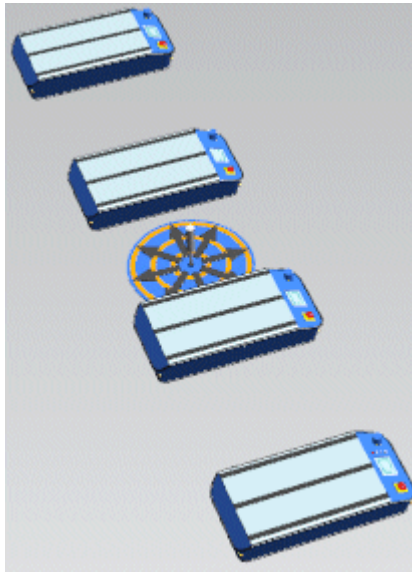
The *AGVPool* looks like this:



The *AGVPool* does initially not show any AGV. To show the AGVs, select the check box **Show Content** on the tab **Graphics** of the *AGVPool*.



Then, click  **Start Simulation**.



Use the object **Marker**  to set way points along which the AGV drives from the *AGVPool* to its destination.

To show a **tooltip** with information about the *AGVPool*, hover with the mouse over it.

To change the length of the graphic and the anchor points of the *AGVPool*, click **Show Manipulators**  on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model

To add the object *AGVPool* to your simulation model, click  **Manage Class Library** > **Basic Objects** > **Resources** > **AGVPool** on the **Home** ribbon tab.

See also

Model an Automated Guided Vehicle System (AGVS)

Dialog Box of the AGVPool

Dialog Box of the AGVPool

Double-click the icon of the *AGVPool* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under **Dialog Items of the Objects**.

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the [Tab Attributes](#).

AGV [text box]

AGV [text box]

Select the vehicle that you want to use for the *Automated Guided Vehicle* (AGV) of the *Automated Guided Vehicle System* (AGVS) in your model.

AGV

*.UserObjects.Transporter...

Remarks

Note

By default, the **Transporter** in the folder **MUs** in the *Class Library* is the AGV. For this reason the methods, attributes, and read-only attributes of the *Transporter* apply to the AGV as well.

Plant Simulation creates the specified **Amount** of AGVs during the **init phase** of the simulation run.

For the AGV that drives freely within the area, you can define these settings for the **Distance Control**:

Length Zone 1, **ΔWidth Zone 1**, **Length Zone 2**, and **ΔWidth Zone 2**

Distance	self.OnDistance...		1	m
Zone 1	Length	1	ΔWidth	0.3
Zone 2	Length	2	ΔWidth	0.6

SimTalk

AGV [SimTalk]

AGVPool [SimTalk] of the Transporter

StoppingCounter [SimTalk] of the Transporter

See also

setRoute [SimTalk] - Transporter

setRouteSegments [SimTalk]

Amount [text box] - AGVPool

Type in the number of vehicles that you want to use for the *Automated Guided Vehicle System* (AGVS).

Amount 

Remarks

Plant Simulation creates the specified **amount** of **AGVs** during the **init phase** of the simulation run.

SimTalk

Amount [SimTalk] - AGVPool

Shift Calendar [AGVPool]

Select the *ShiftCalendar*. It contains the data of the shifts in your installation and controls during which shifts the *AGVPool* works in the plant.

Shift Calendar 

Remarks

Click the ellipsis button and select the *ShiftCalendar* in the dialog **Select Object**.

Or select the *ShiftCalendar* in a *Frame*, drag it to the text box and drop it there. *Plant Simulation* then automatically enters the objects into the **list** of the objects on the tab **Resources** of the *ShiftCalendar*.

You can edit the contents of the text boxes with the commands on the [Context Menu of Embedded Lists](#).

SimTalk

[ShiftCalendarObject \[SimTalk\] - material flow objects](#)

See also

[ShiftCalendar \[object\]](#)

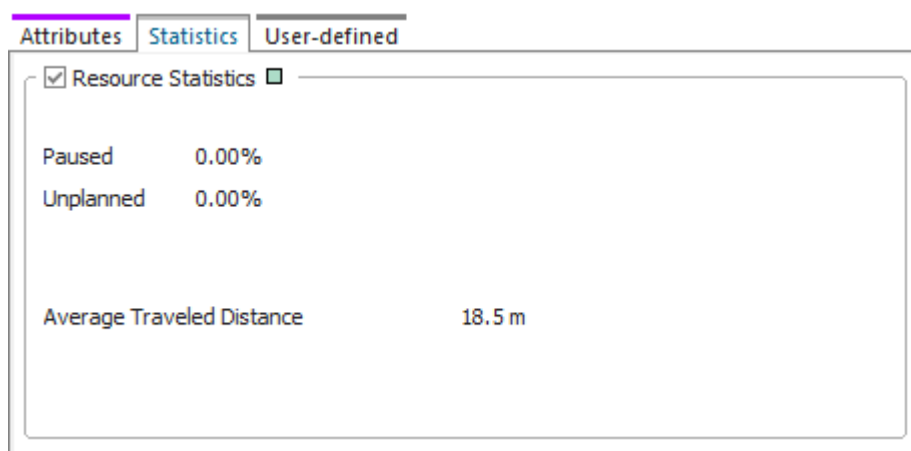
[Select Object \[for controls\]](#)

Tab Statistics [AGVPool]

The tab **Statistics** shows the most important statistics values.

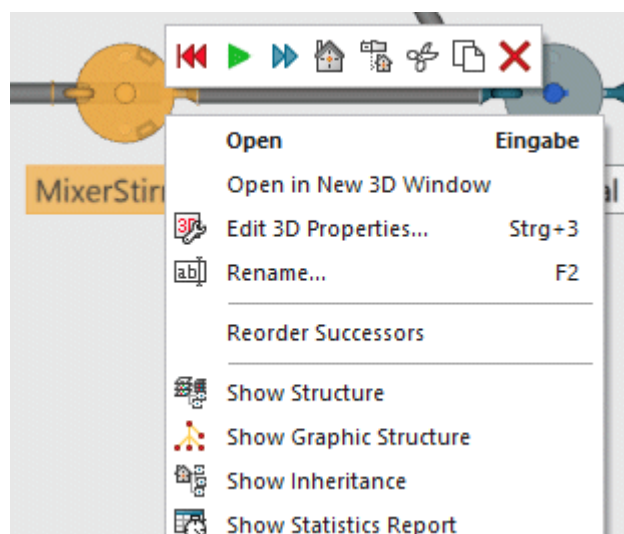
Remarks

Item English	Description	Read-only attribute	Item German
Paused	Shows the portion of the statistics collection period during which the <i>AGVPool</i> was paused.	StatPausingCount [SimTalk]	Pausiert
Unplanned	Shows the portion of the statistics collection period during which the <i>AGVPool</i> was unplanned, i.e., is not scheduled to work to the statistics collection period.	StatUnplannedPortion [SimTalk]	Ungeplant
Average Traveled Distance	Shows the average distance in meters which the AGV traveled from the <i>AGVPool</i> to its destination.	StatAverageTraveledDistance [SimTalk] - AGVPool	Durchschnittlich zurückgelegter Weg



To view [Resource Statistics of Stationary Resources](#) in the [Statistics Report](#), select **View > Show Statistics Report** in the dialog of the object, or click [Show Statistics Report](#) on the **Home** ribbon tab.

You can also click the object in the *Frame* with the right mouse button and select [Show Statistics Report](#) on the context menu.



Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

View Menu

The **View Menu** provides commands to access its functions.

[Refresh \[on View menu\]](#)

[Show Attributes and Methods \[on View menu\]](#)

[Assigned AGVs \[AGVPool\]](#)

[Associated Shift Calendar](#)

[SimTalk](#)

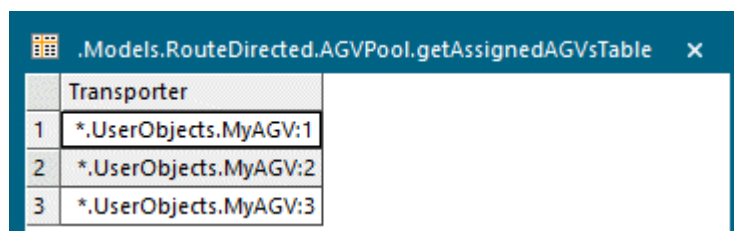
[updateDialog \[SimTalk\]](#)

See also

[View Menu \[general description\]](#)

Assigned AGVs [AGVPool]

Opens a list showing all the AGVs that the *AGVPool* manages.



The screenshot shows a window titled ".Models.RouteDirected.AGVPool.getAssignedAGVsTable" with a close button (X). Inside the window is a table with two columns. The first column is labeled "Transporter" and the second column is labeled "AGV". The table contains three rows of data:

	Transporter	AGV
1	*,UserObjects.MyAGV:1	
2	*,UserObjects.MyAGV:2	
3	*,UserObjects.MyAGV:3	

SimTalk

getAssignedAGVsTable [SimTalk]

Tools Menu

The **Tools Menu** provides these menu commands:

Edit Controls > **Init control**

Edit Observers

Help Menu

The commands are described under the [Help Menu](#).

Methods of the AGVPool

_Methods of the AGVPool

The *AGVPool* provides:

- The *methods* listed in the table of contents to the left.
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			[[Contents:table]] -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				([IconName:string, Width:integer, Height:integer]) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer:=1]) -> object
MUPart	VOID			([Index:integer:=1]) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).